

Angles formed by chords, secants and tangents

Vertex inside, on, or outside the circle — three positions, three rules, one pattern.

LESSON 57

1 × 40 MIN

GEOMETRY 8, TEMA 58

STAGE 9 · 7.1

LESSON CLOCK

5'

12'

12'

7'

4'

Where is the vertex?	5 min
Vertex inside	12 min
Vertex outside	12 min
Tangent and chord	7 min
Homework	4 min

LEARNING OBJECTIVES

- Find the angle between two chords meeting inside a circle.
- Find the angle between two secants meeting outside a circle.
- Use the tangent–chord angle.
- Choose the right rule from the position of the vertex.

TEXTBOOK

National	Tema 58, pp. 138–141
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

One pattern, three rules

Every rule in this lesson measures an angle by the arcs it cuts off. Which rule you use depends only on **where the vertex is**:

VERTEX	ANGLE	NAME
at the centre	$= arc$	central angle
on the circle	$= \frac{1}{2} \cdot arc$	inscribed angle
inside the circle	$= \frac{1}{2} (arc_1 + arc_2)$	two chords
outside the circle	$= \frac{1}{2} (arc_{far} - arc_{near})$	two secants

Reading down the table, the vertex moves outward and the angle shrinks: from the whole arc, to half of it, to half a sum, to half a difference.

TANGENT AND CHORD

The angle between a tangent and a chord drawn from the point of tangency is **half** the arc the chord cuts off — the inscribed-angle rule, with the tangent standing in for a second chord that has shrunk to

nothing.

Using them

Two chords meet inside a circle, cutting arcs of 70° and 30° :

$$\angle = \frac{70^\circ + 30^\circ}{2} = 50^\circ$$

Two secants meet outside, cutting arcs of 100° (far) and 40° (near):

$$\angle = \frac{100^\circ - 40^\circ}{2} = 30^\circ$$

WHICH ARCS?

For the **inside** rule use the two arcs that the vertical angles stand on — the arc “in front” and the arc “behind”. For the **outside** rule always subtract the nearer arc from the further one; the answer must be positive, which is a useful check.

02 Worked examples

EXAMPLE 1 Two chords meet inside a circle, intercepting arcs of 86° and 54° . Find the angle between them.

① The vertex is inside — half the sum.

② $\angle = \frac{86^\circ + 54^\circ}{2}$

③ $= 70^\circ$

And the other pair of vertical angles is 110° .

ANSWER

70°

EXAMPLE 2 Two secants from an external point intercept arcs of 130° and 50° . Find the angle at the point.

① The vertex is outside — half the difference.

② $\angle = \frac{130^\circ - 50^\circ}{2}$

Far arc minus near arc.

③ $= 40^\circ$

Positive ✓

ANSWER

40°

EXAMPLE 3 A tangent touches a circle at T , and a chord TA cuts off an arc of 120° . Find the tangent–chord angle.

① $\angle = \frac{1}{2} \cdot 120^\circ$

Half the intercepted arc.

② $= 60^\circ$

③ On the other side the angle is 120° .

The two make 180° along the tangent line.

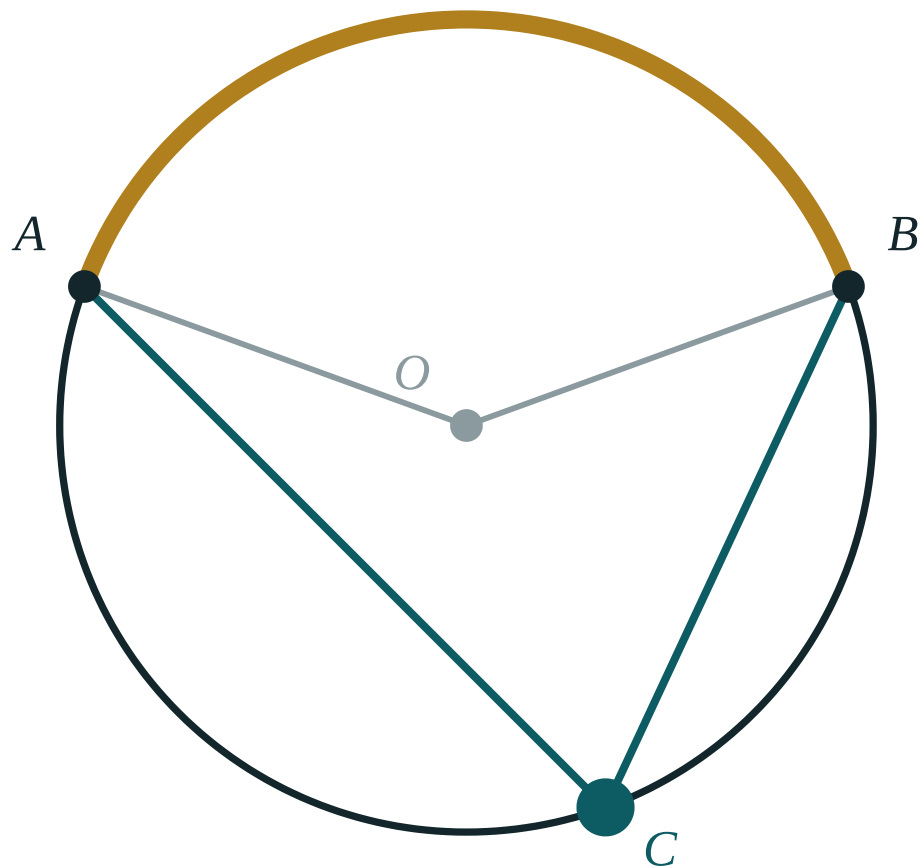
ANSWER

60°

03 Interactive model

Move the points and check each angle against the arcs it intercepts.

Angles and their arcs

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Chord	Vatar	Хорда
Secant	Kesuvchi	Секущая
Tangent–chord angle	Urinma va vatar burchagi	Угол между касательной и хордой
Intercepted arc	Kesilgan yoy	Высекаемая дуга
Vertex inside	Uchi ichkarida	Вершина внутри
Vertex outside	Uchi tashqarida	Вершина снаружи
Half the sum	Yarim yig'indi	Полусумма
Half the difference	Yarim ayirma	Полуразность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two chords meet inside, cutting arcs of 80° and 40° . The angle is:

Two secants meet outside, cutting arcs of 110° and 30° . The angle is:

A tangent–chord angle cuts off an arc of 100° . The angle is:

The vertex is on the circle. The angle is:

the arc

half the arc

half the sum

half the difference

For an external vertex you subtract:

the far arc from the near

the near arc from the far

nothing

both from 360°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Two chords meet inside, arcs 60° and 40° . Find the angle.

ANSWER 50°

2. Two secants meet outside, arcs 100° and 20° . Find the angle.

ANSWER 40°

3. A tangent–chord angle cuts off an arc of 80° . Find it.

ANSWER 40°

4. An inscribed angle stands on a 70° arc. Find it.

ANSWER 35°

5. A central angle stands on a 70° arc. Find it.

ANSWER 70°

6. Two chords meet inside, arcs 90° and 90° . Find the angle.

ANSWER 90°

7. Two secants meet outside, arcs 140° and 60° . Find the angle.

ANSWER 40°

Medium · the standard question

7 PROBLEMS

1. Two chords meet inside, arcs 118° and 46° . Find both pairs of vertical angles.

ANSWER 82° and 98°

2. Two secants meet outside at 25° , and the near arc is 40° . Find the far arc.

ANSWER 90°

3. A tangent–chord angle is 35° . Find the arc and the arc on the other side.

ANSWER 70° and 290°

4. Two chords meet inside at 65° . One arc is 50° . Find the other.

ANSWER 80°

5. Two secants meet outside, far arc 150° , angle 45° . Find the near arc.

ANSWER 60°

6. A tangent and a chord cut off arcs of 140° and 220° . Find both angles at the point of tangency.

ANSWER 70° and 110°

7. Two chords meet inside at 90° . Find the sum of the two intercepted arcs.

ANSWER 180°

Hard · stretch and reason

7 PROBLEMS

1. Two secants from P cut a circle so that the arcs, in order, are 100° , 60° , 140° and 60° . Find $\angle P$.

ANSWER $\frac{140^\circ - 60^\circ}{2} = 40^\circ$

2. A tangent at T and a secant from the same external point cut off arcs of 160° and 60° . Find the angle at that point.

ANSWER 50°

3. Two chords AB and CD meet at P inside a circle, with arc $AC = 74^\circ$ and arc $BD = 36^\circ$. Find $\angle APC$.

ANSWER 55°

4. An angle at an external point is 30° and the two arcs differ by how much?

ANSWER 60°

5. A quadrilateral is inscribed in a circle and one exterior angle is 85° . Find the interior angle opposite to it.

ANSWER 85° — the exterior angle equals the opposite interior angle

6. A tangent–chord angle equals the inscribed angle on the other side of the chord. Explain why.

ANSWER Both equal half the same intercepted arc — the tangent behaves like a chord that has shrunk to a point.

7. Two secants meet outside at 20° . The near arc is 3 times smaller than the far arc. Find both arcs.

ANSWER 60° and 20°

07 Homework

Homework — 5 problems

Geometry 8, Tema 58, pp. 138–141. Write down where the vertex is before choosing the rule.

1. *Two chords meet inside, arcs 96° and 44° . Find the angle.*
2. *Two secants meet outside, arcs 150° and 50° . Find the angle.*
3. *A tangent–chord angle cuts off an arc of 130° . Find both angles at the point of tangency.*
4. *Two chords meet inside at 72° . One arc is 60° . Find the other.*
5. *Two secants meet outside at 35° , near arc 45° . Find the far arc.*